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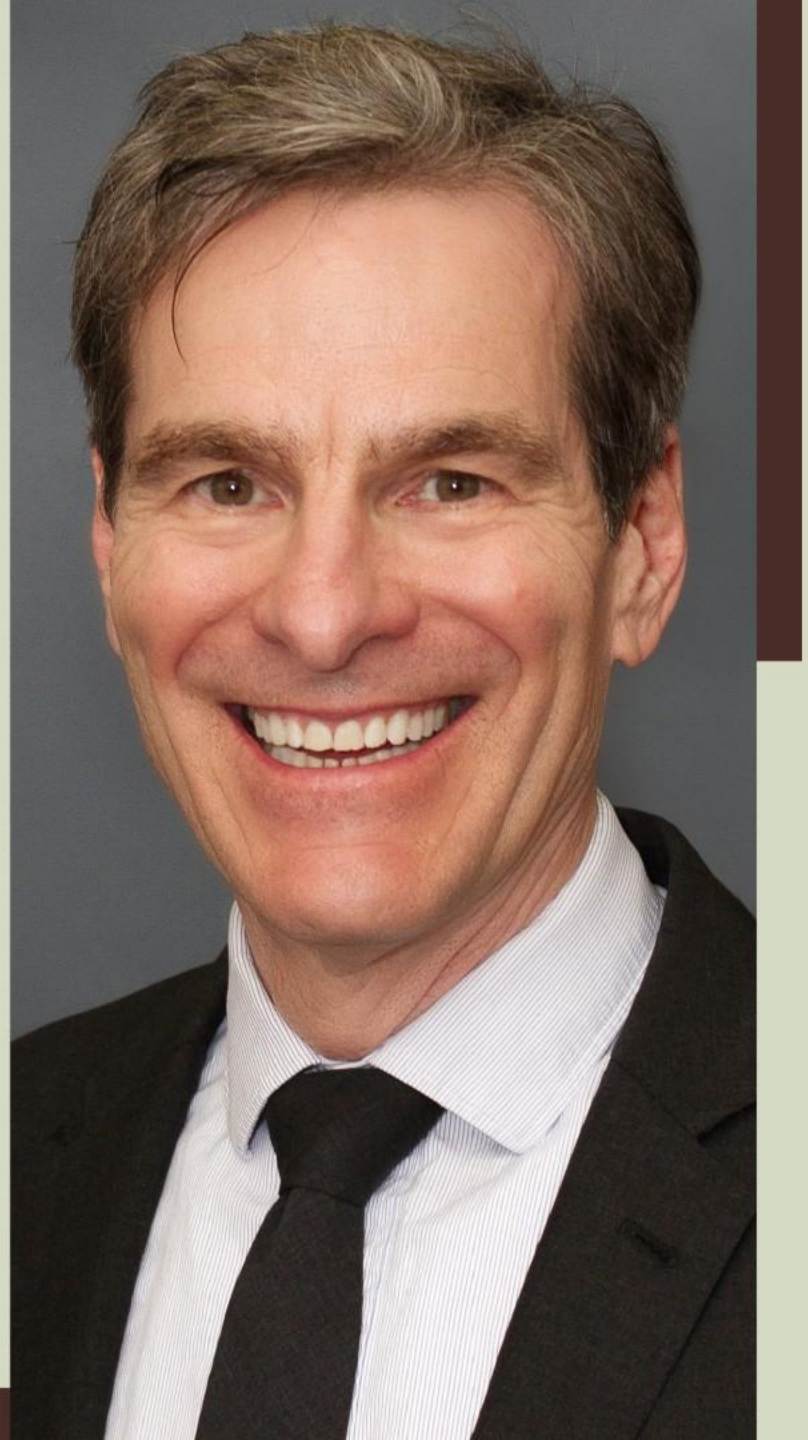
KOF

Principles of Macroeconomics

Lecture 2: National Accounting

Prof. Dr. Jan-Egbert Sturm



SCHWEIZERISCHE NATIONALBANK
BANQUE NATIONALE SUISSE
BANCA NAZIONALE SVIZZERA
BANCA NAZIUNALA SVIZRA
SWISS NATIONAL BANK

Karl Brunner Distinguished Lecture

"Inflation" by John H. Cochrane,
Hoover Institution (Stanford University)

Thursday, 2 October 2025, ETH Zurich, Audi Max (HG F 30)
Drinks reception: 4.45 – 5.30 pm; Lecture: 5.30 – 7.00 pm

Welcome by Joël Mesot
President of ETH Zurich

Introduction by Martin Schlegel
Chairman of the Governing Board, Swiss National Bank

Moderated by Tanya König

Live streaming www.snb.ch, Research TV

Please note that there will be a security check at the entrance to the lecture hall. Please arrive well in advance of the start of the event and refrain from bringing bags that are larger than A4 (21 x 30 cm).

Hosted by **KOF** Swiss Economic Institute and Chair of Macroeconomics: Innovation and Policy

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«Welt aus den Fugen – Stresstest für die Schweiz? Risiken erkennen, Resilienz stärken, Zukunft sichern»

KOF Prognosetagung
25. September 2025, ab 12:30 Uhr

ExpertInnen:
Sermîn Faki, Nachrichtenchefin CH Media
Prof. Dr. Hans Gersbach, Co-Direktor KOF, ETH Zürich
Anne Schwöbel, Mitglied Verwaltungsrat, B. Braun Medical AG
Prof. Dr. Jan-Egbert Sturm, Direktor KOF, ETH Zürich

Mehr Infos

Schedule

Date	Lecture	Topic	Chapters
16/09/2025	1	Introduction	1, 2
23/09/2025	2	National Accounting	20
30/09/2025	3	Production and Growth	21
07/10/2025	4	Unemployment and the labour market	22
14/10/2025	5	Saving, investment and the financial system	23
21/10/2025	6	The Monetary System	24
28/10/2025	7	International Trade	3,15,17
04/11/2025	8	Open-Economy Macroeconomics	25
11/11/2025	9	Business Cycles	26
18/11/2025	10	Keynesian Economics and IS-LM Analysis	27
25/11/2025	11	Aggregate Demand and Aggregate Supply	28
02/12/2025	12	Monetary and Fiscal Policy	29
09/12/2025	13	Phillips Curve / Supply-side policies	30,31
16/12/2025	14	Exam preparation	

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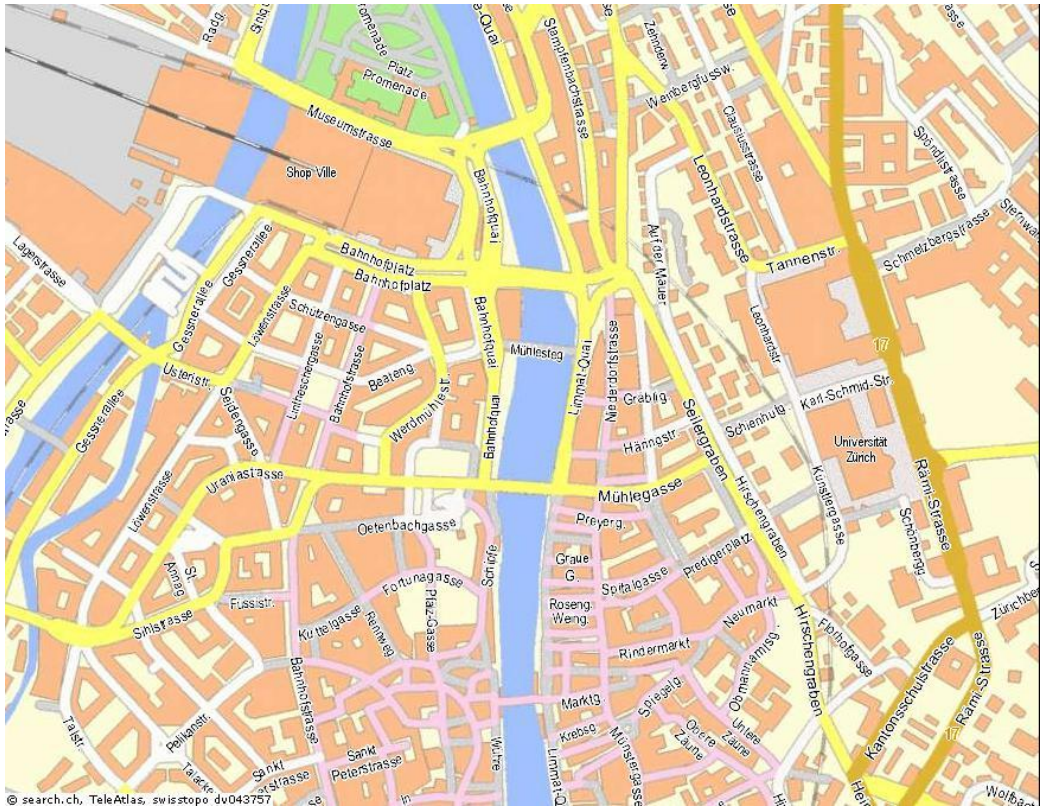
Principles of Macroeconomics

Lecture 1: Introduction

Prof. Dr. Jan-Egbert Sturm

The Role of Assumptions

- Economists make assumptions in order to make the world easier to understand
- The art in scientific thinking is deciding which assumptions to make
- Economists use different assumptions to answer different questions



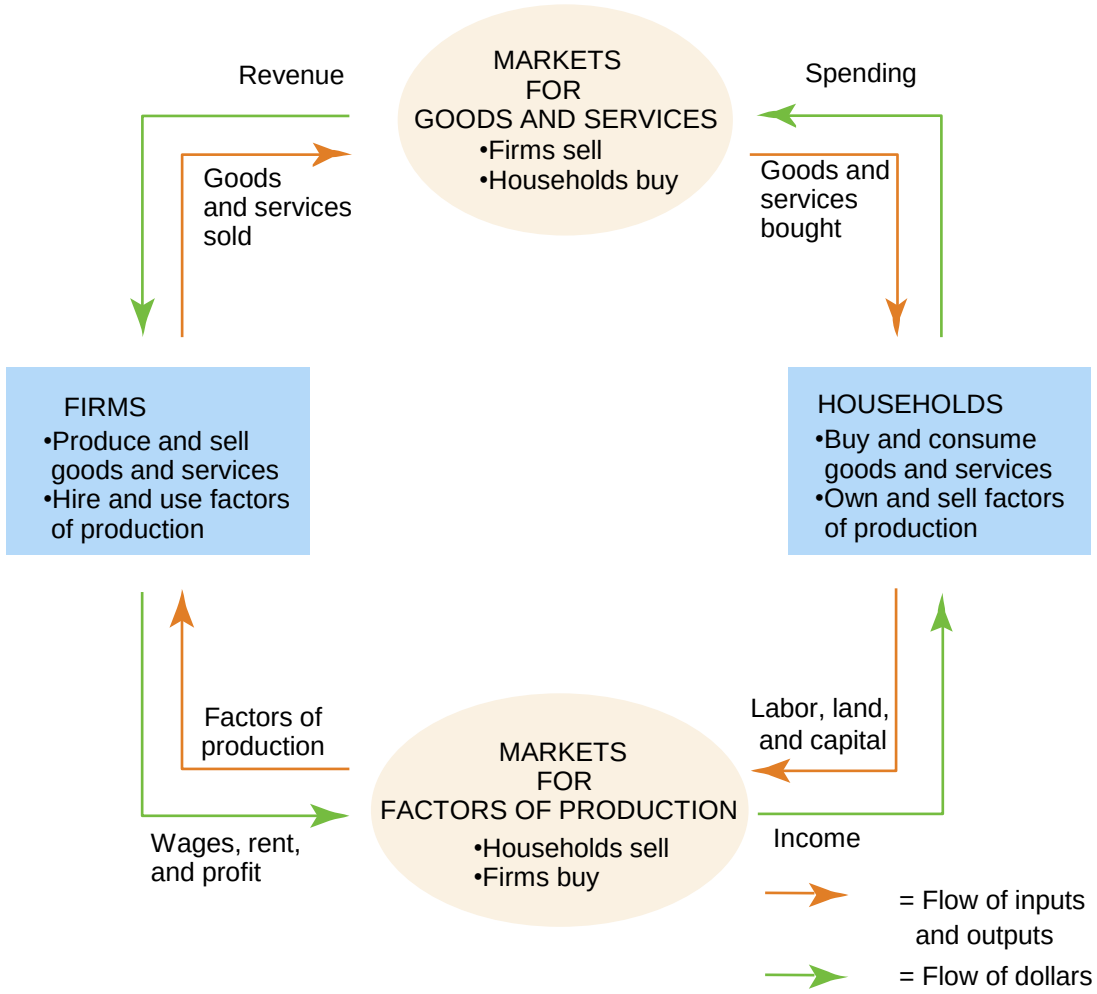
Which will be more useful on your next trip to London?

Understanding complex systems requires abstraction

Economic Models

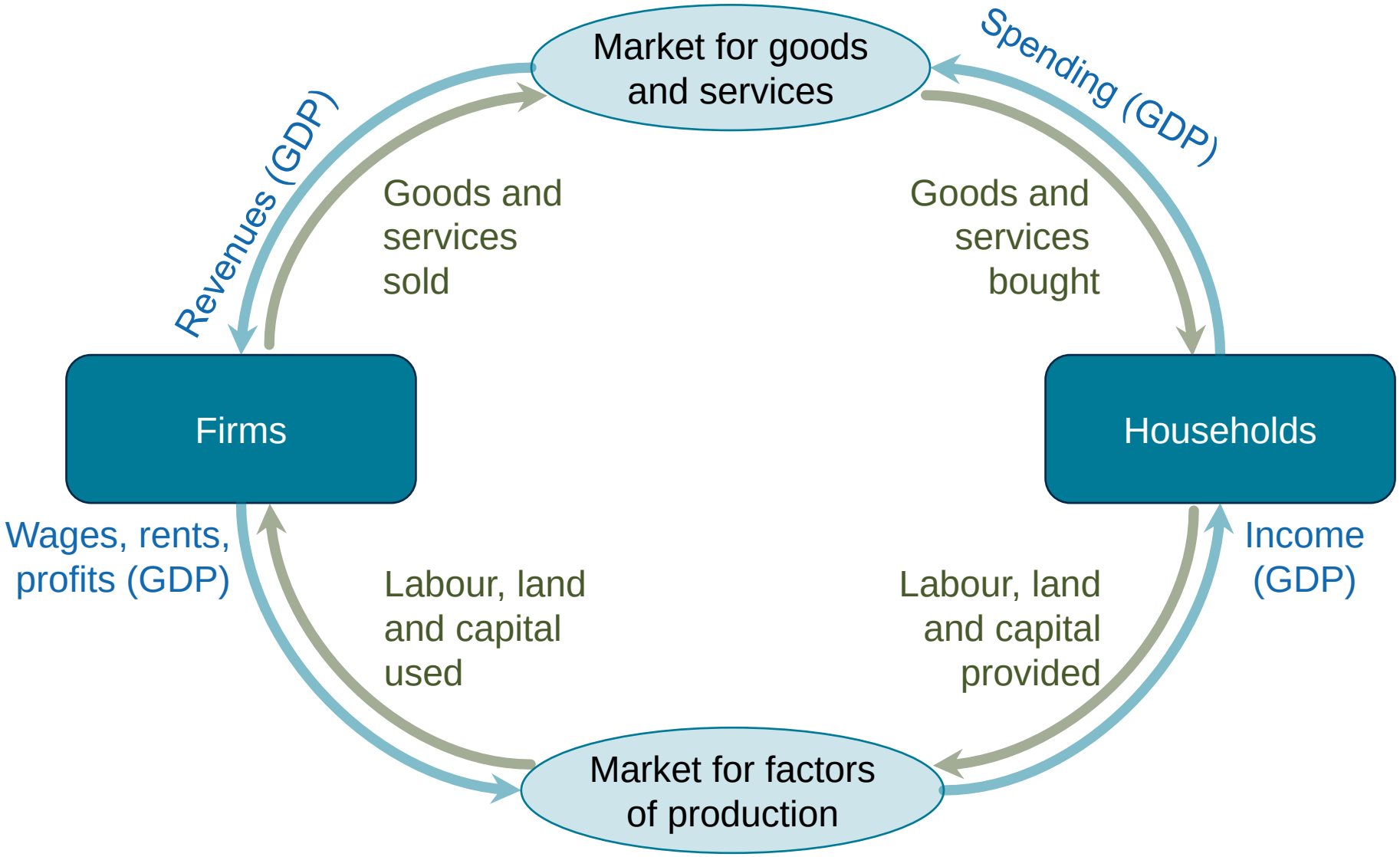
- Economists use models to simplify reality in order to improve our understanding of the world
- A model will contain a number of variables.
  - Definition of an endogenous variable:
    - a variable whose value is determined within the model.
  - Definition of an exogenous variable:
    - a variable whose value is determined outside the model.
  - Cause and effect
    - exogenous variables determine endogenous variables
  - Ceteris paribus meaning ‘other things equal’, is used to note when other factors that might affect outcomes are assumed to be constant
- Models are more unstable the longer the time period being considered and forecast
- Two of the most basic economic models are:
  - The Circular Flow Diagram
  - The Production Possibilities Frontier

Our First Model: The Circular-Flow Diagram

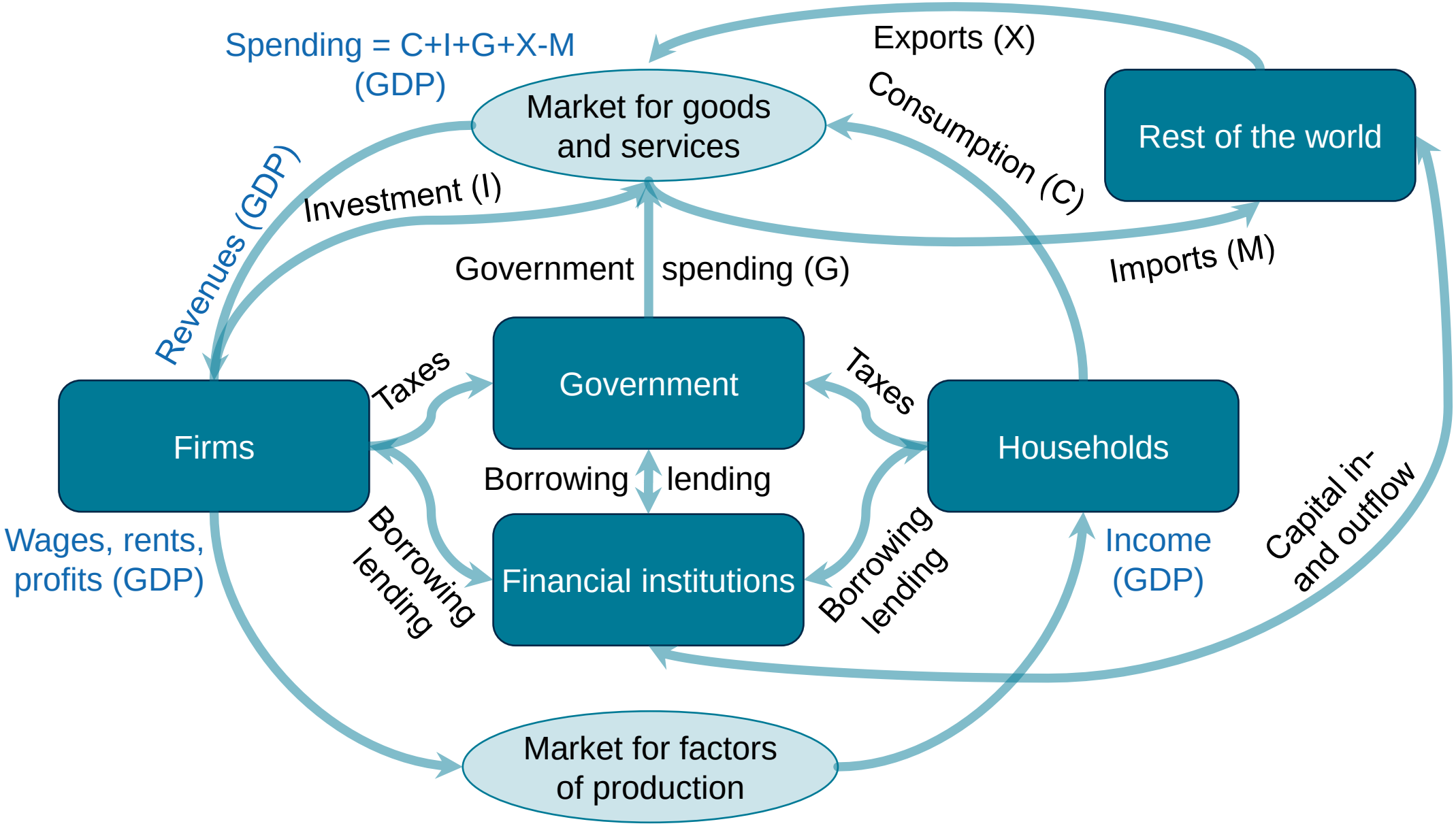


- The ***circular-flow diagram*** is a visual model of the economy that shows how dollars flow through markets among households and firms
  - Factors of Production
    - Inputs used to produce goods and services
    - Land, labour, and capital
  - Markets for Factors of Production
    - Households sell / Firms buy
  - Markets for Goods and Services
    - Firms sell / Households buy
  - Households
    - Buy and consume goods and services
    - Own and sell factors of production
  - Firms
    - Produce and sell goods and services
    - Hire and use factors of production

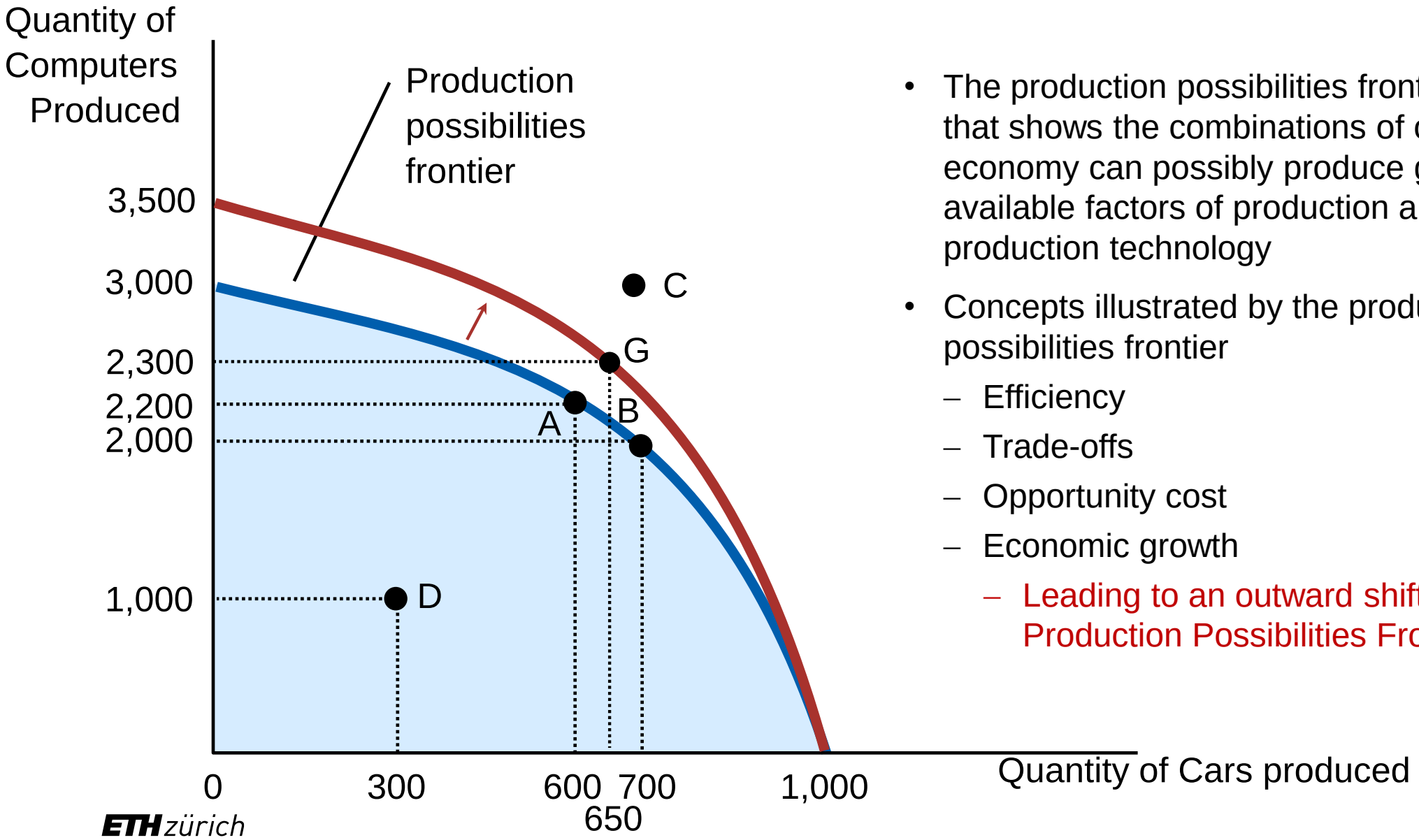
A circular-flow diagram of the economy



Financial flows in a circular-flow diagram in a more realistic economy



Our Second Model: The Production Possibilities Frontier



- The production possibilities frontier is a graph that shows the combinations of output that the economy can possibly produce given the available factors of production and the available production technology
- Concepts illustrated by the production possibilities frontier
  - Efficiency
  - Trade-offs
  - Opportunity cost
  - Economic growth
    - Leading to an outward shift of the Production Possibilities Frontier

THE ECONOMIST AS POLICY ADVISOR

- When economists are trying to explain the world, they are **scientists**
- When economists are trying to change the world, they are **policy advisors**

Positive versus Normative Analysis

- *Positive statements* are statements that attempt to describe the world as it is
  - Called descriptive analysis

- *Normative statements* are statements about how the world should be
  - Called prescriptive analysis

Positive versus Normative Analysis

?

- Are the following positive or normative statements?
  - An increase in the minimum wage will cause a decrease in employment among the least-skilled
  - POSITIVE

?

  - Higher federal budget deficits will cause interest rates to increase
  - POSITIVE

?

  - The income gains from a higher minimum wage are worth more than any slight reductions in employment
  - NORMATIVE

  - Governments should be allowed to collect from tobacco companies the costs of treating smoking-related illnesses among the poor
  - NORMATIVE

?

WHY ECONOMISTS DISAGREE

- They may disagree about the validity of alternative positive theories about how the world works
- They may have different values and, therefore, different normative views about what policy should try to accomplish

Do Economist Agree? (KOF-NZZ Economists Survey, December 2024)

https://kof.ethz.ch/en/surveys/experts-surveys/economists-surveys/econstat.html

- Guaranteed property rights and free competition are central to increasing a country's material well-being
  - 92.6% agree – 3.4% disagree
- Without publicly funded basic research, there is too little innovation and growth
  - 92.5% agree – 5.1% disagree
- Tariffs and import quotas reduce the material well-being in a country
  - 80.8% agree – 9% disagree
- Wage and/or price controls should be used as a tool to fight inflation
  - 6.8% agree – 78.4% disagree
- Taxing emissions is a better method to tackle pollution than setting limits
  - 77.8% agree – 10.7% disagree
- Climate-neutral growth will be possible with new technologies
  - 70.5% agree – 11.9% disagree
- Rent control reduces the quantity or quality of housing supply
  - 69.7% agree – 13.7% disagree
- The government expenditure share of GDP in Switzerland is too high
  - 13.1% agree – 65.3% disagree
- Disposable income in Switzerland should be distributed more evenly
  - 39.6% agree – 34.5% disagree

Summary (1)

- The principles of personal decision making are:
  - People face trade-offs.
  - The cost of something is what you give up to get it.
  - Rational people think at the margin.
  - People respond to incentives.
- The principles of economic interaction are:
  - Trade can make everyone better off.
  - Markets are usually a good way to organize economic activity.
  - Governments can sometimes improve market outcomes (if there is some market failure or if the market outcome is inequitable).
- The principles of the economy as a whole are:
  - A country's standard of living depends on its ability to produce goods and services.
  - Prices rise when the government prints too much money.
  - Society faces a short-run trade-off between inflation and unemployment.

Summary (2)

- Economics is divided into two subfields:
  - Microeconomics is the study of decision-making by households and firms in the marketplace.
  - Macroeconomics is the study of the forces and trends that affect the economy as a whole.
- A positive statement is an assertion about how the world is.
- A normative statement is an assertion about how the world ought to be.
- When economists make normative statements, they are acting more as policy advisors than scientists.
- Economists who advise policymakers offer conflicting advice either because of differences in scientific judgments or because of differences in values.
- At other times, economists are quite united in the advice they offer, but policymakers may choose to ignore it.

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SFSO Video: What have statistics ever done for us?

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Recapitulate the basic division in economics:

- *Microeconomics* is the study of how individual households and firms make decisions and how they interact with one another in markets.

- *Macroeconomics* is the study of the *economy as a whole*. Its goal is to explain the economic changes that affect many households, firms, and markets at once.
  - Macroeconomics tries to answer questions like the following:
    - Why is average income high in some countries and low in others?
    - Why do prices rise rapidly in some time periods while they are more stable in others?
    - Why do production and employment expand in some years and contract in others?

MEASURING A NATION'S WELL-BEING

- When judging whether the economy is doing well or poorly, it is natural to look at the total income that everyone in the economy is earning.

- For an economy as a whole, income must equal expenditure because:
  - Every transaction has a buyer and a seller.
  - Every dollar of spending by some buyer is a dollar of income for some seller.

- The equality of income and expenditure can be illustrated with the circular-flow diagram.

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- *Gross domestic product* (GDP) is a measure of the income and expenditures of an economy.

- Because for the total economy income equals expenditure,
  we can calculate GDP by adding up total expenditure in the economy,
  or by adding up total income in the economy.

- **GDP is the total market value of all final goods and services produced within a country in a given period of time.**

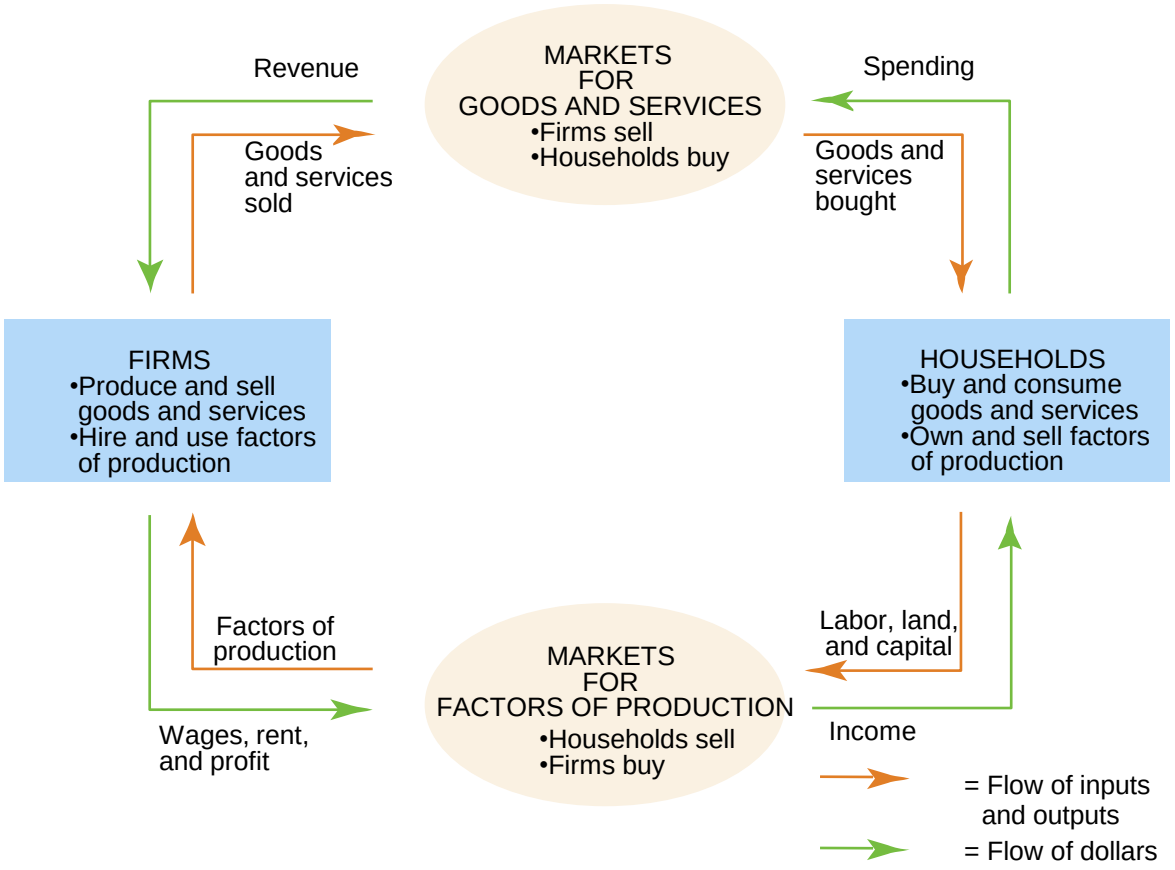
THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- “GDP is the Market Value . . .”
  - Output is valued at market prices.
- “. . . Of All. . .”
  - Includes all items produced in the economy and legally sold in markets
    - What Is Not Counted in GDP?
      - GDP excludes most items that are produced and consumed at home and that never enter the marketplace.
      - It often excludes items produced and sold illicitly, such as illegal drugs.
- “. . . Final . . .”
  - It records only the value of final goods and services, not intermediate goods (the value is counted only once).

THE MEASUREMENT OF GROSS DOMESTIC PRODUCT

- “. . . Goods and Services . . .”
  - It includes both tangible goods (food, clothing, cars) and intangible services (haircuts, housecleaning, doctor visits).
- “. . . Produced . . .”
  - It includes goods and services currently produced, not transactions involving goods produced in the past.
- “ . . . Within a Country . . .”
  - It measures the value of production within the geographic confines of a country.
- “. . . In a Given Period of Time.”
  - It measures the value of production that takes place within a specific interval of time, usually a year or a quarter (three months).

How to measure GDP? Three main approaches

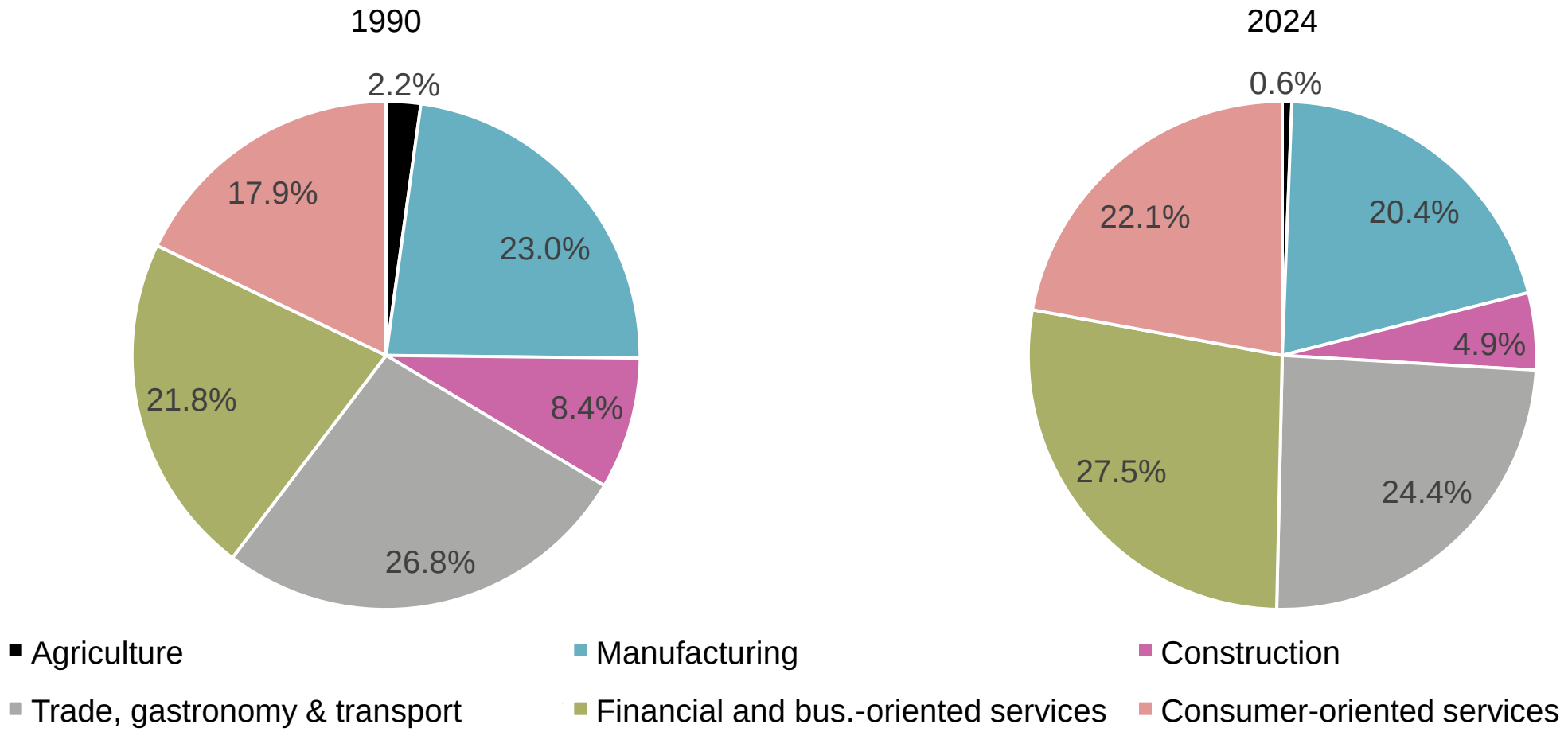


1. Expenditure Approach (demand side)
   – GDP = consumption + investment + government spending + net exports

2. Production Approach (supply side)
   – GDP = ∑(Gross Output− Intermediates)

3. Income Approach (income side)
   – GDP = Wages+Rents+Interest+Profits

Swiss GDP from the production (supply) side



Source: SFSO

DEMAND-SIDE COMPONENTS OF GDP (Y)

- Consumption (C)
  - The spending by households on goods and services, with the exception of purchases of new housing
- Investment (I)
  - The spending on capital equipment, inventories, and structures, including new housing
- Government Purchases (G)
  - The spending on goods and services by local, state, and federal governments
    - Does not include transfer payments, as these are not made in exchange for currently produced goods or services
- Net Exports (NX = X-M)
  - Exports (X) minus imports (M)

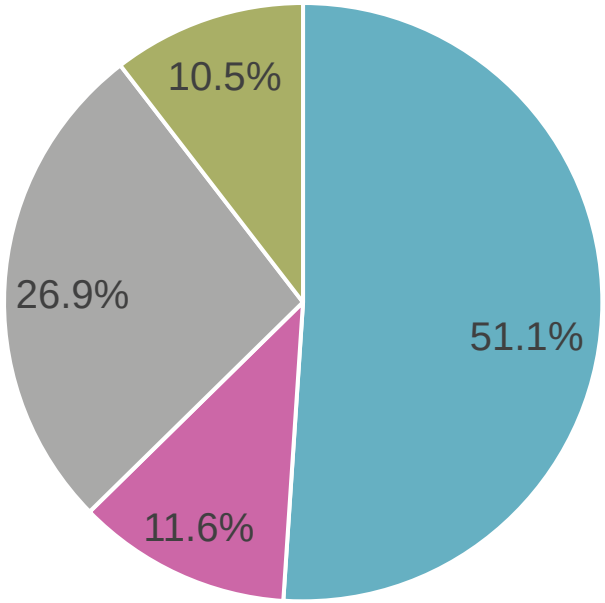
$$Y = C + I + G + NX$$

$$= C^{dom} + C^{for} + I^{dom} + I^{for} + G^{dom} + G^{for} + X - (C^{for} + I^{for} + G^{for})$$

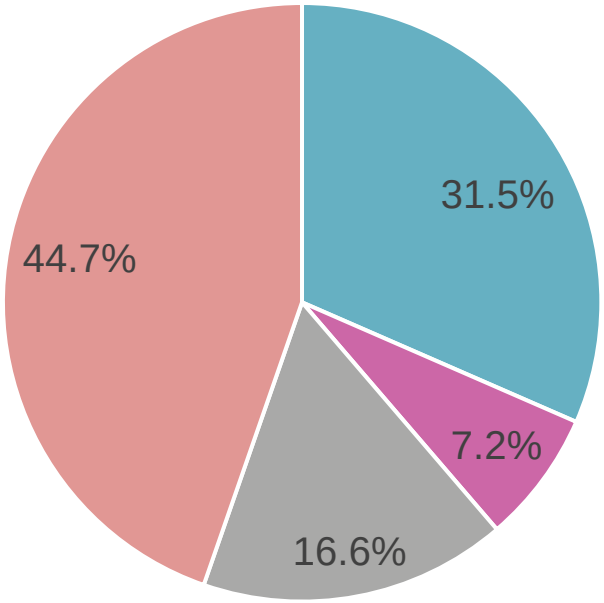
$$= C^{dom} + I^{dom} + G^{dom} + X$$

GDP and its demand components (Swiss GDP in 2024: 826 Bln CHF (in prices of 2023))

Gross Domestic Product:
$Y = C + G + I + (X-M)$



Total demand: $D = C + G + I + X$
$D = (C^{dom}+C^{for}) + (G^{dom}+G^{for}) + (I^{dom}+I^{for}) + X$



C = Private consumption | G = Public consumption | I = Investment | NX = Net exports | X = Exports

Source: SFSO

(nominal) GDP and Its Components: Switzerland

	Total (in Mln. CHF)	Per person (in CHF)	Percent of total
	2024		
GDP, Y	825,796	91,687	100%
Consumption, C	421,716	46,823	51%
Investment, I	223,114	24,772	27%
Government consumption, G	95,813	10,638	12%
Net exports, NX = X-M	85,153	9,454	10%
Exports, X	596,944	66,278	72%
Imports, M	511,791	56,824	62%

Source: SFSO

REAL VERSUS NOMINAL GDP

- If total spending rises from one year to the next, this can be because...
  - ...the economy is producing a larger output of goods and services *and/or*
  - ...goods and services are being sold at higher prices.
- We want a measure of the total output of goods and services that is not affected by changes in the prices of those goods and services.

- *Nominal GDP* values the production of goods and services at *current prices*.
- *Real GDP* values the production of goods and services at *constant prices*.

- Real GDP is a more accurate measure of economic well-being than nominal GDP, because changes in real GDP are not affected by changes in prices, but reflect only changes in production.

Real and Nominal GDP

Prices and Quantities				
Year	Price of Hot dogs	Quantity of Hot dogs	Price of Hamburgers	Quantity of Hamburgers
2005	$1	100	$2	50
2006	$2	150	$3	100
2007	$3	200	$4	150

Calculating Nominal GDP	
2005	($1 per hot dog × 100 hot dogs) + ($2 per hamburger × 50 hamburgers) = $200
2006	($2 per hot dog × 150 hot dogs) + ($3 per hamburger × 100 hamburgers) = $600
2007	($3 per hot dog × 200 hot dogs) + ($4 per hamburger × 150 hamburgers) = $1,200

Calculating Real GDP (base year 2005)	
2005	($1 per hot dog × 100 hot dogs) + ($2 per hamburger × 50 hamburgers) = $200
2006	($1 per hot dog × 150 hot dogs) + ($2 per hamburger × 100 hamburgers) = $350
2007	($1 per hot dog × 200 hot dogs) + ($2 per hamburger × 150 hamburgers) = $500

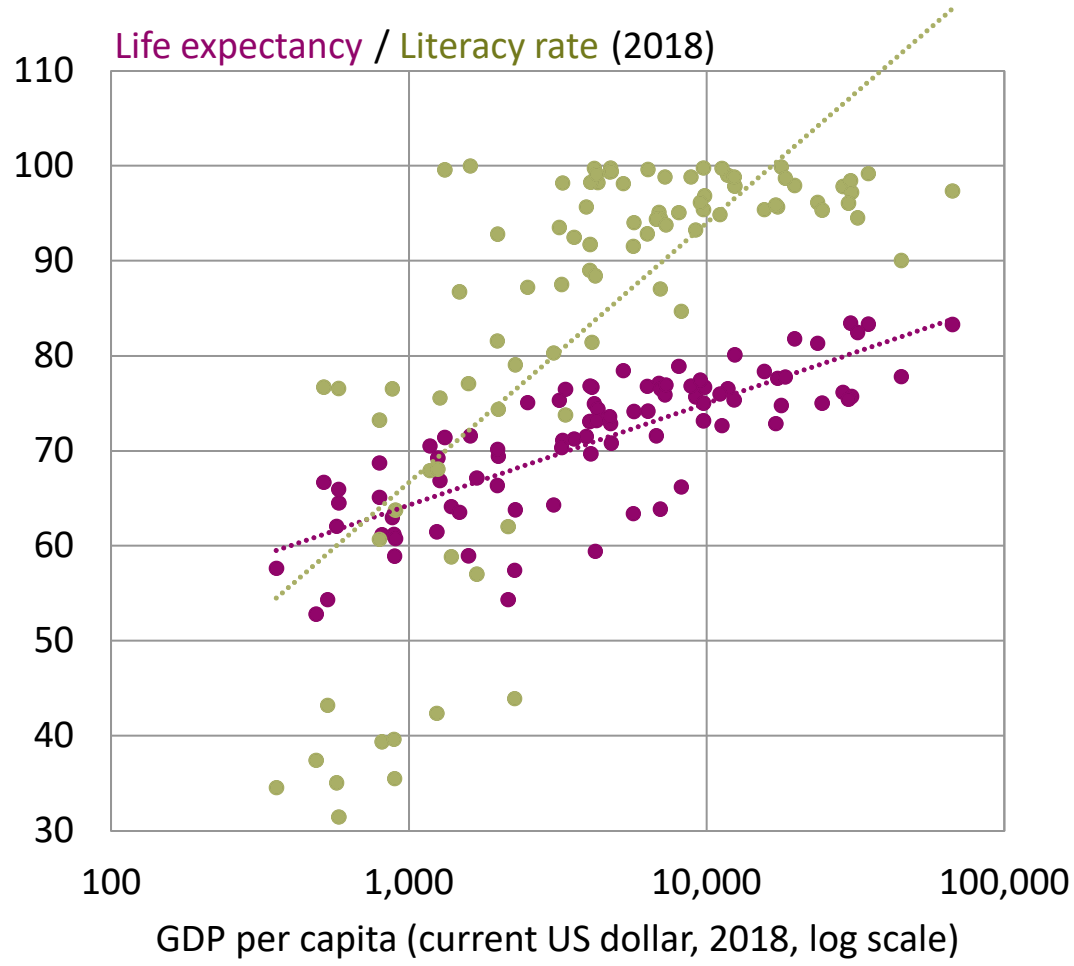
Measurement Issues in Practice

- In most western countries the statistical offices produce a single measure of GDP and they do this by combining income, expenditure and output approaches.
  - In theory these measures should produce the same result (in practice they don't).

- There are various practical difficulties in measuring GDP.
  - Under-reporting in order to avoid tax
  - Changes in the goods and services produced over time *render the use of base years problematic*.

- Annual chain linking can be used to tackle this latter problem.
  - Rather than updating the base year every five or ten years,
    this method does it every year, calculating the prices in previous years' prices.

IS GDP A GOOD MEASURE OF ECONOMIC WELL-BEING?



Sources: IMF, World Bank

- GDP is the best single measure of economic well-being
- GDP per person tells us the income and expenditure of the average person in the economy
- Higher GDP per person indicates a higher standard of living
- GDP is not a perfect measure of happiness or quality of life
  - See article on moodle (here)
- Some things that contribute to well-being are not included
  - The value of leisure
  - The value of a clean environment
  - The distribution of income
  - The value of almost all activity that takes place outside of markets, such as the value of the time parents spend with their children and the value of volunteer work
- Income, life expectancy and CO2 emission (video):
- The MONET 2030 indicator system gives an overview of sustainable development in Switzerland and goes beyond GDP

MEASURING THE COST OF LIVING

- The *inflation rate* is the percentage change in the price level from the previous period.

  Formally: $$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} * 100$$

- *Inflation* refers to a situation in which the economy's overall price level is rising.

- *Deflation* refers to situation in which the economy's overall price level is falling.

The GDP Deflator

- The *GDP deflator* is a measure of the price level calculated as the ratio of nominal GDP to real GDP times 100.
- It tells us what portion of the rise in nominal GDP is attributable to a rise in prices rather than a rise in the quantities produced.
- The GDP deflator is calculated as follows

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

	Calculating the GDP Deflator
2005	($200/$200) × 100 = 100
2006	($600/$350) ×100 = 171
2007	($1,200/$500) × 100 = 240

THE CONSUMER PRICE INDEX

- The *consumer price index (CPI)* is a measure of the overall cost of the goods and services bought by a typical consumer.

- It is used to monitor changes in the cost of living over time.

- When the CPI rises, the typical consumer has to spend more money to maintain the same standard of living.

How the Consumer Price Index Is Calculated

1. *Fix the basket*: Determine what prices are most important to the typical consumer.
   - Identify a market basket of goods and services the typical consumer buys.
   - Consumer surveys to set the weights for the prices of those goods and services.
2. *Find the prices*: Find the prices of each of the goods and services in the basket for each point in time.
3. *Compute the basket's cost*: Use the data on prices to calculate the cost of the basket of goods and services at different times.
4. *Choose a base year and compute the index:*
   - Designate one year as the base year, making it the benchmark against which other years are compared.
   - Compute the index by dividing the price of the basket in one year by the price of the basket in the base year.

$$\text{Consumer price index in year t} = \frac{\text{Price of basket in year t}}{\text{Price of basket in base year}} \times 100$$

5. *Compute the consumer price inflation rate:* The *consumer price inflation rate* is the percentage change in the price level – as measured by CPI – from the preceding period.

$$\text{Inflation rate in year 2} = \frac{\text{CPI in year 2 - CPI in year 1}}{\text{CPI in year 1}} \times 100$$

Calculating Consumer Price Index and the Inflation Rate: An Example

Step 1: Survey Consumers to Determine a Fixed Basket of Goods

Basket = 4 hot dogs, 2 hamburgers

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot Dogs	Price of Hamburgers
2005	$1	$2
2006	2	3
2007	3	4

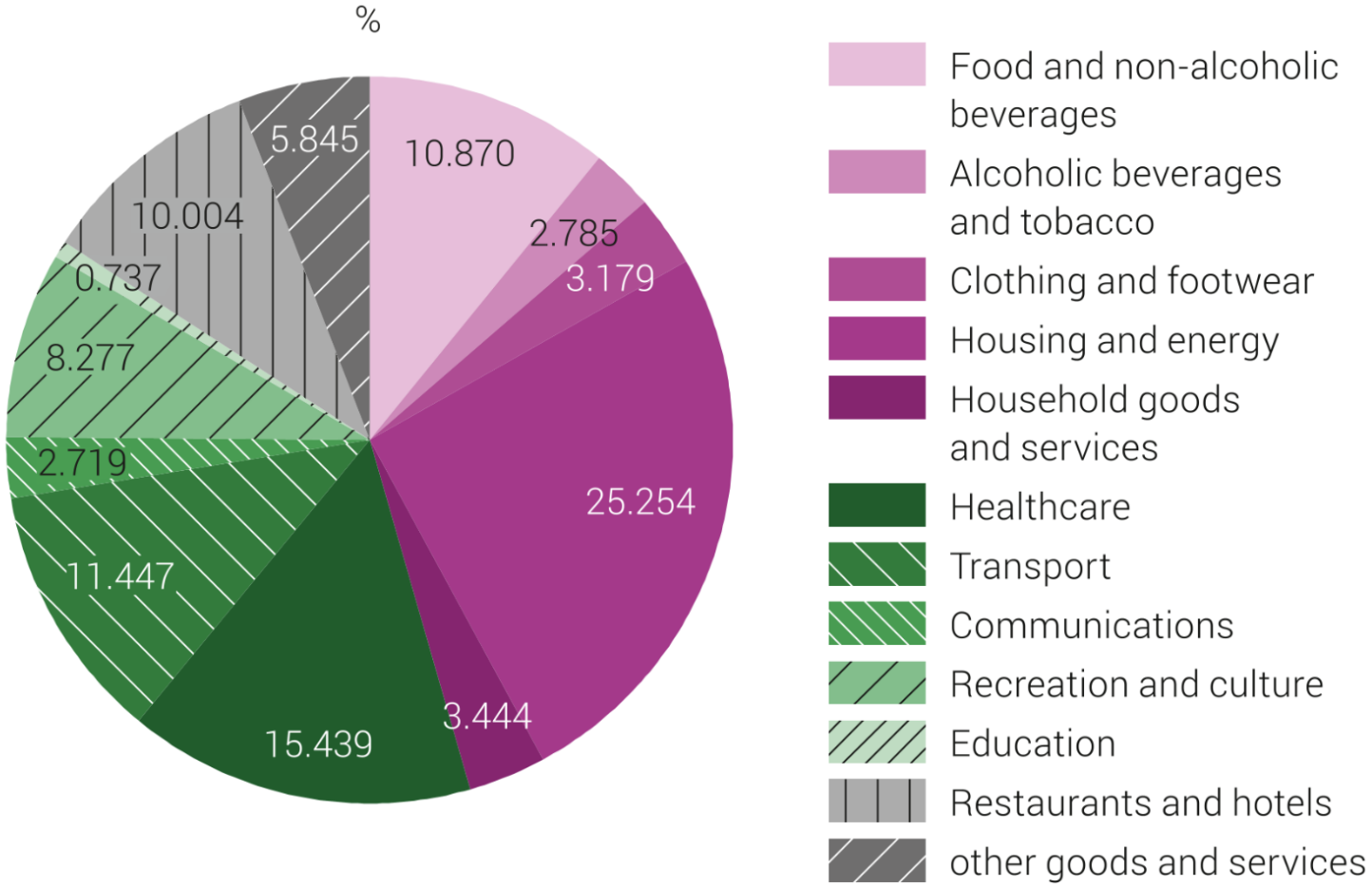
Calculating Consumer Price Index and the Inflation Rate: An Example

Step 3: Compute the Cost of the Basket of Goods in Each Year	
2005	($1 per hot dog × 4 hot dogs) + ($2 per hamburger × 2 hamburgers) = $8 per basket
2006	($2 per hot dog × 4 hot dogs) + ($3 per hamburger × 2 hamburgers) = $14 per basket
2007	($3 per hot dog × 4 hot dogs) + ($4 per hamburger × 2 hamburgers) = $20 per basket

Step 4: Choose One Year as a Base Year (2005) and Compute the Consumer Price Index in Each Year	
2005	($8/$8) × 100 = 100
2006	($14/$8) × 100 = 175
2007	($20/$8) × 100 = 250

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year	
2006	(175 − 100)/100 × 100 = 75%
2007	(250 − 175)/175 × 100 = 43%

CPI basket and weights in Switzerland, 2024



Historical developments in RPI (≈CPI) weights for the United States

RPI GROUP	1987	2012
Food	167	114
Catering	46	47
Alcohol	76	56
Tobacco	38	29
Housing	157	237
Fuel and light	61	46
Household goods	73	62
Household services	44	67
Clothing and footwear	74	45
Personal goods and services	38	39
Motoring expenditure	127	131
Fares and other travel costs	22	23
Leisure goods	47	33
Leisure services	30	71

Source: SFSO

SFSO videos on the most important price statistics in Switzerland

- Price statistics in brief
  - https://www.youtube.com/watch?v=xZdopkGNdeo

- The Consumer Price Index (CPI) at a glance
  - https://www.youtube.com/watch?v=t2BXjbiLmMo

- How does the consumer price index work?
  - https://www.youtube.com/watch?v=lhmjO-gvo9s

- The Producer Price Index (PPI) at a glance
  - https://www.youtube.com/watch?v=7FvQW583sew

Problems in Measuring the Cost of Living

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.
- There are three key issues that cause the CPI to overstate the true cost of living:
  1. Substitution bias
  2. Introduction of new goods
  3. Unmeasured quality changes

Problems in Measuring the Cost of Living

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.
- There are three key issues that cause the CPI to overstate the true cost of living:
  1. Substitution bias
     - The basket does not change to reflect consumer reaction to changes in relative prices.
       - Consumers substitute toward goods that have become relatively less expensive.
       - The index overstates the increase in cost of living by not considering consumer substitution.
  2. Introduction of new goods
  3. Unmeasured quality changes

Problems in Measuring the Cost of Living

- The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.
- There are three key issues that cause the CPI to overstate the true cost of living:
  1. Substitution bias
  2. Introduction of new goods
     - The basket does not reflect the change in purchasing power brought on by the introduction of new products.
       - New products result in greater variety, which in turn makes each euro more valuable.
       - Consumers need less money to maintain any given standard of living.
  3. Unmeasured quality changes

Problems in Measuring the Cost of Living

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  2. Introduction of new goods
  3. Unmeasured quality changes
     - If the quality of a good rises from one year to the next, the value of a euro rises, even if the price of the good stays the same.
       - If the quality of a good falls from one year to the next, the value of a euro falls, even if the price of the good stays the same.
       - The statistical office tries to adjust the price for constant quality, but such differences are hard to measure.

Problems in Measuring the Cost of Living

- The issue over accurate measurement is important because many government programs use the CPI to adjust for changes in the overall level of prices.

- The statistical office calculates other prices indexes:
  - The *producer price index*, which measures the cost of a basket of goods and services bought by firms rather than consumers.
    - Because firms eventually pass on higher costs to consumers in the form of higher prices on products, the producer price index is believed to be helpful in predicting changes in the CPI.

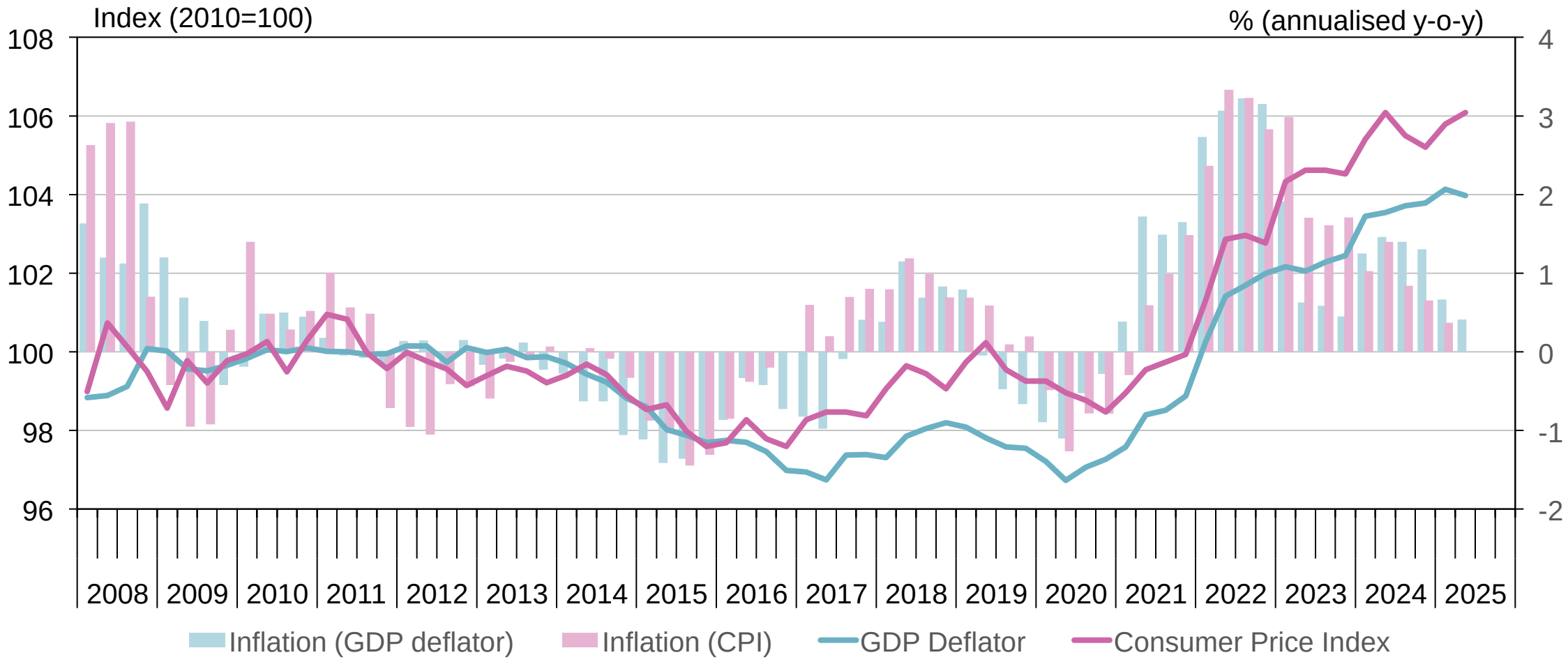
The GDP Deflator versus the Consumer Price Index

- Remember, the GDP deflator is calculated as follows:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

- Economists and policymakers monitor both the GDP deflator and the consumer price index to gauge how quickly prices are rising.
- There are two important differences between the indexes that can cause them to diverge.
  1. The *GDP deflator* reflects the prices of all goods and services *produced domestically*, whereas... …the *consumer price index* reflects the prices of all goods and services *bought by consumers*.
  2. The *consumer price index* compares the price of a *fixed basket* of goods and services to the price of the basket in the base year (only every five years does the SFSO change the basket)... …whereas the *GDP deflator* compares the price of *currently produced* goods and services to the price of the same goods and services in the base year.

Two measures of inflation in Switzerland



Sources: SFSO, Seco

CORRECTING ECONOMIC VARIABLES FOR THE EFFECTS OF INFLATION

- Price indexes are used to correct for the effects of inflation when comparing money figures from different times.
- Example:
  - In 1947 MPs earned £1,000
  - Assume in 2020 that MPs earned £74,000 but prices have risen since 1947.
  - CPI index in 1947 = 28.9
  - CPI index in 2020 = 1,018.6
- Do the following to convert (inflate) MPs' salary in 1947 to a figure in 2020 values:

$$\text{Salary}_{2020} = \text{Salary}_{1947} \frac{\text{Price level in 2020}}{\text{Price level in 1947}}$$

$$= £1{,}000 \frac{1{,}018.6}{28.9} = £35{,}246$$

Indexation

- When some money amount is automatically corrected for inflation by law or contract, the amount is said to be *indexed* for inflation.
  - Many government transfer payment systems use indexation for the benefits.
    The government often also adjusts tax brackets used for income tax in line with inflation.
  - There are uses of indexation in the private sector as well.
    Many contracts between firms and trade unions include Cost-of-Living Allowances.

Real and Nominal Interest Rates

- Interest represents a payment in the future for a transfer of money in the past.
- The *nominal interest* rate is the interest rate usually reported and not corrected for inflation.
  - It is the interest rate that a bank pays.
- The *real interest rate* is the nominal interest rate that is corrected for the effects of inflation.
- Here is a simple example
  - You borrowed €1,000 for one year.
  - Nominal interest rate was 15%.
  - During the year inflation was 10%.

$$\textit{Real interest rate} = \textit{Nominal interest rate} - \textit{Inflation}$$

$$= 15\% - 10\% = 5\%$$

SUMMARY

- Because every transaction has a buyer and a seller, the total expenditure in the economy must equal the total income in the economy.
- Gross domestic product (GDP) measures an economy's total expenditure on newly produced goods and services and the total income earned from the production of these goods and services.
- GDP is the market value of all final goods and services produced within a country in a given period of time.
- GDP is divided among four components of expenditure: consumption, investment, government purchases, and net exports.
- Nominal GDP uses current prices to value the economy's production. Real GDP uses constant base-year prices to value the economy's production of goods and services.
- The GDP deflator—calculated from the ratio of nominal to real GDP—measures the level of prices in the economy.
- GDP is a good measure of economic well-being because people prefer higher to lower incomes.
- It is not a perfect measure of well-being because some things, such as leisure time and a clean environment, are not measured by GDP.

SUMMARY

- The consumer price index shows the cost of a basket of goods and services in one year relative to the cost of the same basket in the base year.
- The index is used to measure the overall level of prices in the economy.
- The percentage change in the CPI measures the inflation rate.
- The consumer price index is an imperfect measure of the cost of living for the following three reasons:
  - substitution bias / the introduction of new goods / and unmeasured changes in quality.
- The GDP deflator differs from the CPI because it includes goods and services produced rather than goods and services consumed.
- In addition, the CPI uses a fixed basket of goods, while the GDP deflator automatically changes the group of goods and services over time as the composition of GDP changes.
- Money figures from different points in time do not represent a valid comparison of purchasing power.
- Various laws and private contracts use price indexes to correct for the effects of inflation.
- The real interest rate equals the nominal interest rate minus the rate of inflation.